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We provide sensor and video solution for improving financial performance in transportation

Document

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Company Background

- Canadian founder and investors
- Incorporated in 2021
- Winner of Panasonic Deep Tech Innovation Challenge
- Seeking to develop the product for commercialization

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Project Objective: address financial management pain points in the transportation industry

Problem Statement	Value Proposition
Inefficient data integration processes.	Enhanced Data Integration
Difficulty in accurate revenue forecasting.	Improved Financial Management
Manual processes leading to time and resource drain.	Streamlined Automation
Lack of actionable insights for pricing optimization.	Basic AI Insights
Limited access to real-time operational data.	Real-time Operational Data
High training time and adoption rates.	User-Friendly Interface
Excessive administrative and operational expenses.	Initial Cost Reduction

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Project Charter

- Project Outcomes
- Timeline
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- Field Testing
- Budget

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Project Charter: Project Outcomes

- MVP / Technical Prototype / Field Test
- Commercial prototype
- Further agreement with technology partner

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Project Charter: Timeline

- Timeline:
  - MVP: March or April to August (3 months):
    - Data collection
    - Demonstrate value proposition
    - ‘Sellable’ Product
  - Commercial Deployment: September to December
  - Scaling: December onwards

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Project Charter:

Core Offering	Non-core Offering
• Application • Vehicle IoT System / Collection Point System • AI/ML	• Drive recorder camera recording (with/without geotagging) • Vehicle position tracking • Sensor data logging (e.g., accelerometer, IR) • Operating times • Passenger counting • Data transfer to cloud via cellular/Wi-Fi/mmWave
Field Testing	
• Vehicle type: bus, electric bus • Data: road camera(s), in-vehicle camera(s), telematics, operational/financial data (e.g., fuel consumption, ticket prices) • Upload frequency: daily • Amount of data per upload: 1 bus with 6 standard definition (SD) cameras will generate 50 GB per day (24 hours) • **Upload speed: New Technology can upload 50GB in ~3 minutes @ 2 Gbps (7x faster than alternative)**	

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Project Charter: - Application

- Hosted on the cloud
- Migration of
- Server side communication
- Add IoT data ingestion
- Add ML/AI Integration

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Project Charter: – wireless and telematics tech

- Two ‘boxes’: one for each vehicle and another at collection point
- Data:
  - Telematics
  - In-vehicle camera recordings (door-facing)
    -facing camera recordings
  - Operational and financial data (e.g., fuel)
  - In-vehicle camera recordings (safety and security)
  - Bus driver-facing and front-facing camera recordings
- Assembly

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Project Charter: Field Testing

- Vehicle type: bus, electric bus
- Data collected: road camera(s), in-vehicle camera(s), telematics, operational/financial data (e.g., fuel consumption, ticket prices), third party data (weather, traffic, etc.)
- Route: TBD
- Duration: TBD
- Upload frequency: Daily and Real-time
- Data volume: 1 bus with 6 standard definition (SD) cameras will generate 50 GB per day (24 hours)
- **Upload speed: New Technology can upload 50GB in ~3 minutes @ 2 Gbps (7x faster than alternative)**

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Project Charter: Budget

- Development (Cost to be confirmed):
    1. UX/UI Designer: overall app and user experience
    2. Full-stack developer: server to user data transfer, app interface, database, AI/ML interface, data preparation, ETL, data engineering
    3. IoT engineer: field to server data transfer (including vehicle connection and device assembly)
    4. AI/ML: server-side application development for using data from vehicles, data preparation, ETL, data engineering (find potential insights to sell such as acceleration vs fuel consumption, safety, etc.)
- Hardware components: $5k - $10k (to be confirmed with technology partner)
- Testing facility with vehicle (cost to be confirmed): Faculty Researcher in Transportation department + 2-3 assistants (for vehicle operations)

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